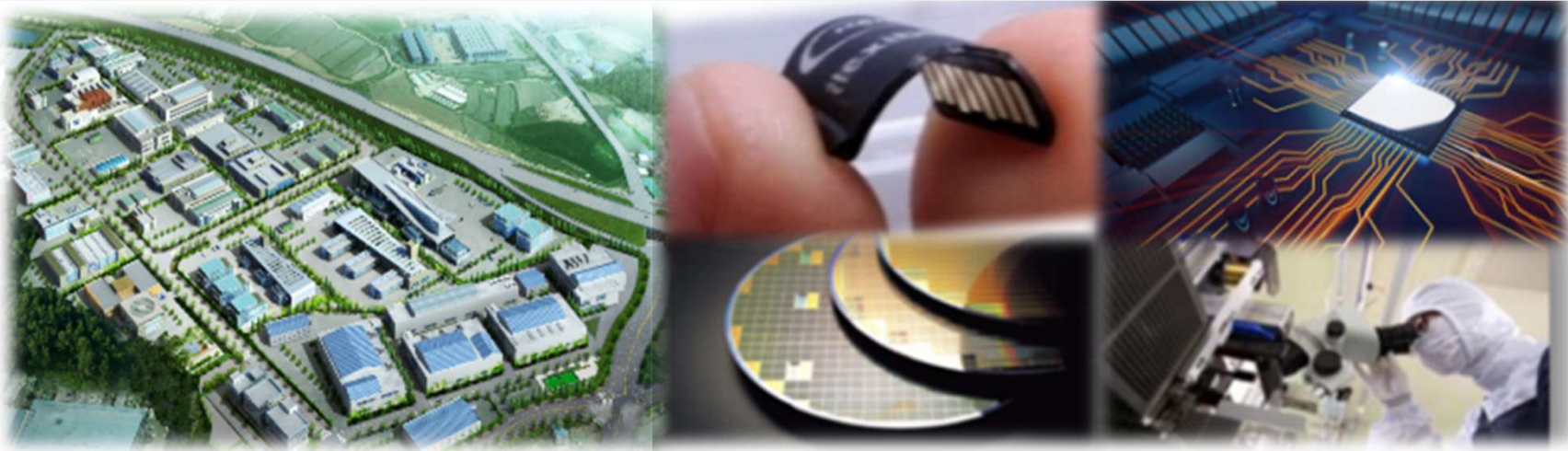




SATURN DOE Report

[DP Process]



2026.09.12

김용모

SATURN Project _ DOE Report



✓ **Purpose** SATURN 제품의 양산성 검증을 위해 DP 공정(Lamination, Back Grind, Saw)의 표준 작업 조건 및 품질 특성 (CTQ)의 공정 능력(Cpk)을 최종 검증함

✓ **Results**

Step	Machine	Raw Material (각 단위 공정에서 선정한 BOM)	Test Item (단위 공정의 주요 평가항목/CTQ)	Result
SMT	HYBRID5	WS693CPS (Alphametal) 0603 Resistor dummy (MURATA) 0402 Capacitor dummy (MURATA)	VISUAL, Capacitor Shear Test	PASS Fail
Lamination	RU-3520	BG Tape: E-6142S (LINTEC)	VISUAL, TAPE VOID	PASS Fail
Back Grind	DGP8761	Rough/Fine Wheel, DAF: FH-7221S(T)	WAFER THICKNESS, ROUGHNESS, TTV	PASS Fail
Wafer Saw	DFD6361	Blade: ZH05-SD3500-N1-50-CE / FD	KERF WIDTH, CHIPPING, CRACK	PASS Fail
DA				
WB				
MD				
BE				

✓ **Conclusion**

2차 DOE 치핑 불량에 대한 원인 분석 및 개선 대책 수립을 완료하여 3차 DOE에서 안정적인 품질을 확보함. DP 전 단위 공정의 CTQ 규격을 만족하였으므로 양산화 검증(Mass Qual) 진행을 승인

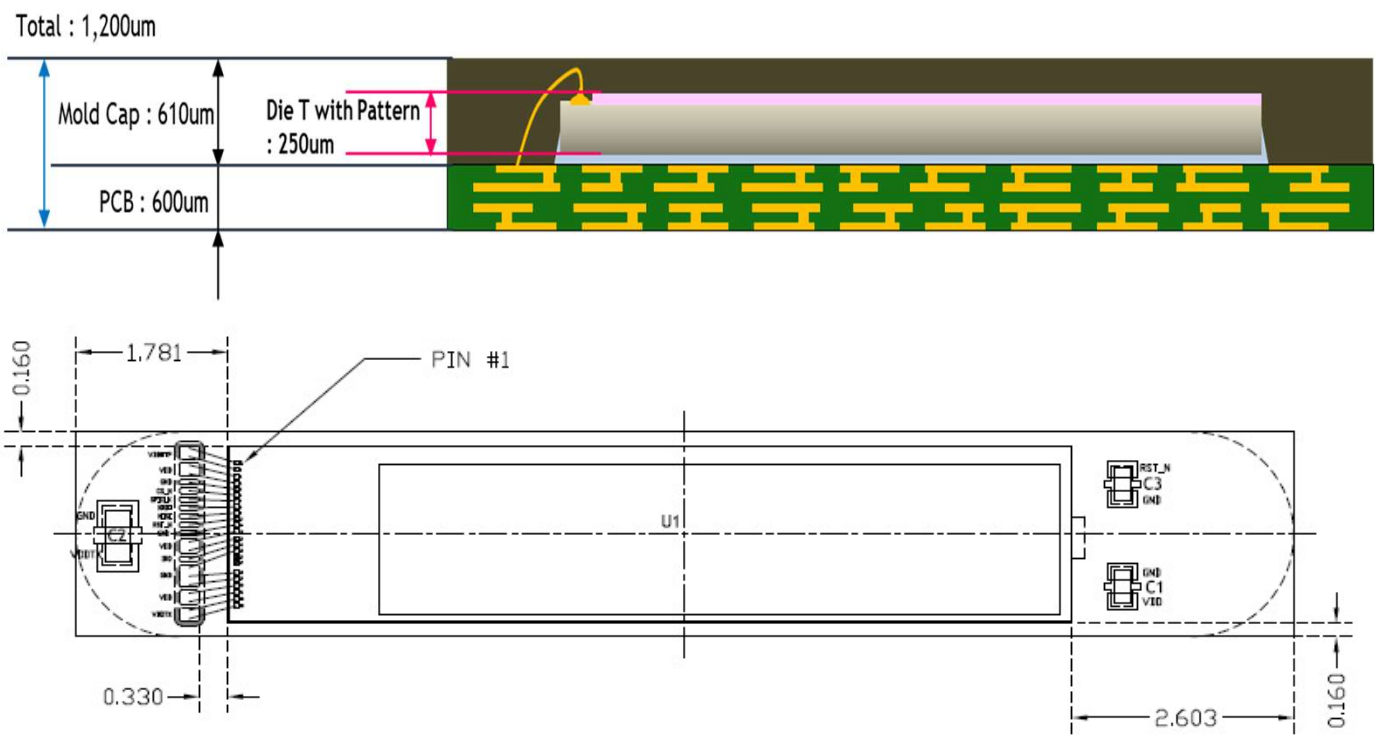
SATURN Project _ DOE Report

✓ Package / Wafer Information

PKG Type	Project Code	Wafer Size(inch)	Die Size(mm)	Target Thickness(um)	Scribe Lane(μm)	DAF Thickness(um)
BGA	SATURN	12	9.916 x 2.080	250	80	20

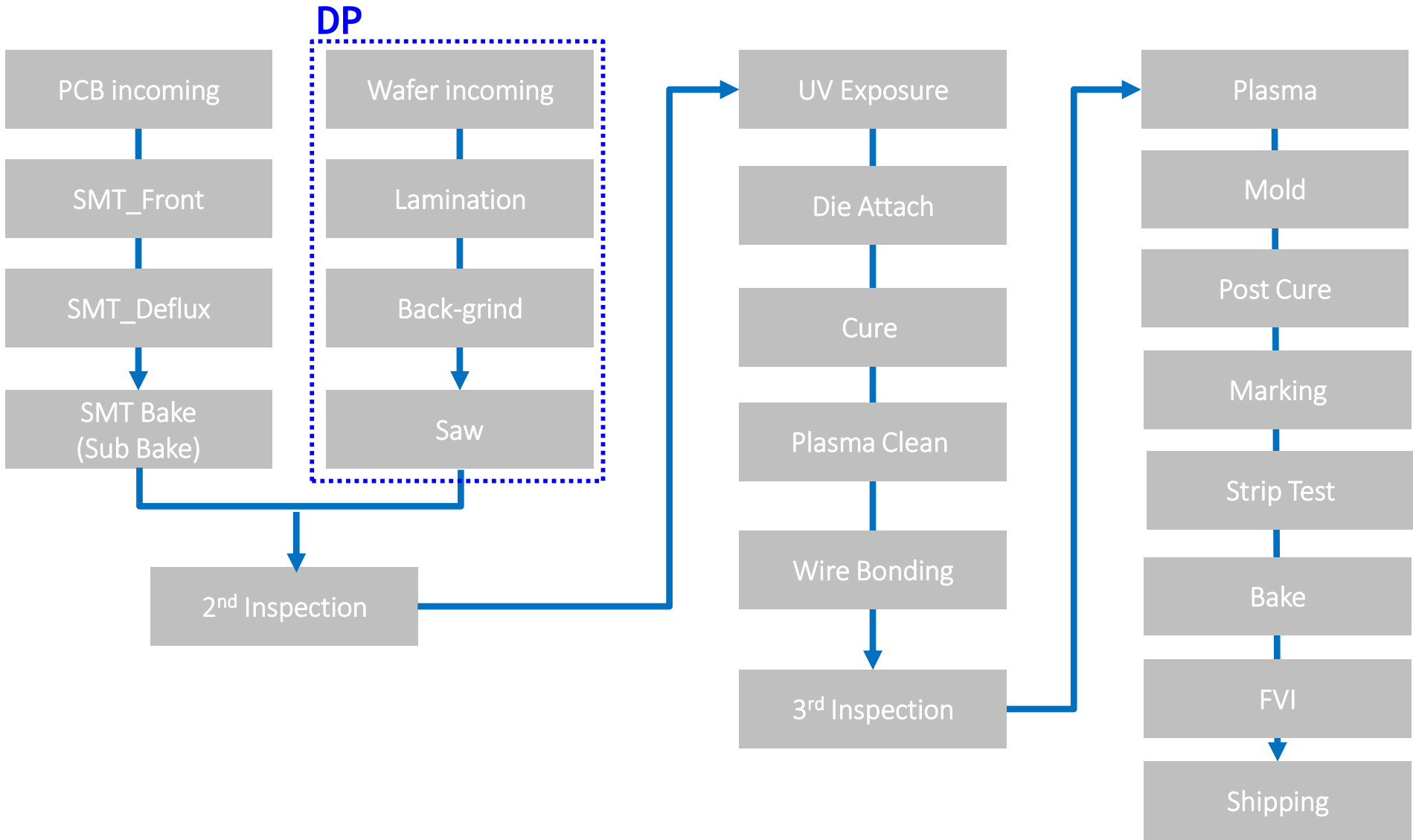
✓ PCB Description

PCB / Substrate			
Spec. No.	Part No.	Core	SR
ADS*****	APF*****	*****	*****



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✓ Process Flow



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✓ 품질 검사 검토 항목 (CTQ Check)

Process	Inspection Item	Sample Size
BG Tape LAMINATION	VISUAL	1 SUB / M.C / 1st LOT
Wafer BACK GRIND	WAFER THICKNESS	13 Points / All Wafer
	WAFER ROUGHNESS	5 Points / Each Wafer
	WAFER TTV	12inch
Wafer Saw	KERF WIDTH	10 Points / All Wafer (또는 10Points / Each Wafer)
	CHIPPING / CRACK	All Wafer / Run (또는 2 Wafer / Each Evaluation)
PCB SMT	VISUAL	1SUB /M.C/1st LOT
DIE ATTACH	VISUAL	1 SUB/M.C/1st LOT
	TAPE VOID	2 UNIT/M.C/1st LOT
	DIE ACCURACY	20 UNIT/M.C/1st LOT
	BONDLINE THICKNESS	20 UNIT/M.C/1st LOT
	PARAMETER	M.C/ 1st LOT

Process	Inspection Item	Sample Size
WIRE BONDING	VISUAL/PARA	1 SUB/M.C/1st LOT
	BOND PULL TEST(*)	55 WIRE/M.C/1st LOT
	BALL SHEAR TEST(*)	22 BALL/M.C/1st LOT
	LOOP HEIGHT	11 WIRE/M.C/1st LOT
	BALL HEIGHT	11 WIRE/M.C/1st LOT
	BALL DIAMETER	11 WIRE/M.C/1st LOT
	KOH ETCHING	5WIRE/M.C/ 3 LOTS
IVI GATE	VISUAL	AQL 0.065%/LOT
MOLDING	VISUAL/PARA	1 SHOT/M.C/1st LOT
	S.A.T	1 SHOT/M.C/1st LOT
	X-RAY	1 SHOT/M.C/1st LOT
	Mold Thickness check	1 STRIP /1st LOT
MARKING	VISUAL	1SUB /M.C/1st LOT

★ HINT ★

초기 제품 개발 검토 단계에서 수립된 검사 항목 및 샘플링 검사 수량을 기입한다.

어떤 목적으로 검토되었던 내용인지 명확히 알고 있어야 하며, 6Page 부터는 검사된 결과가 Data로 명시되어 고객에게 품질 특이사항이 없음을 보여주는 근거 자료로 활용된다.

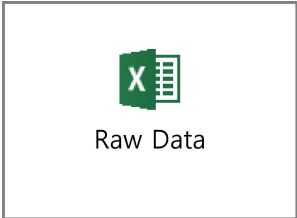
SATURN Project _ DOE Report

1. Back Grind Process Condition & Result

1) Process Condition

Item	Spec	Applied Condition
Mount Table temp (° C)	75 ± 5	75 (또는 75 °C)
Wheel RPM (RPM)	G1 : 2700±500 / G2 : 2700±500	G1 : 2700 / G2 : 2700
Wheel Feed (um/sec)	G1 : 6±4 / 5±3 / 4±3 G2 : 0.7±0.3 / 0.4±0.3 / 0.4±0.3	G1 : 5.0 / G2 : 0.5
Chuck Speed (RPM)	G1 : 300 / 300 / 300 G2 : -300 / -300 / -300	G1 : 300 / G2 : -300

※ Raw Data (참고)



2) Visual Inspection Result (9Page 참조 / Visual 결과는 특이사항 없는 것으로 함)

Item	Lot No.	Sample Size	Inspection Results	
Visual Inspection	DOE Lot	2 wafer / each evaluation	Pass / Fail	2 wafer / each evaluation

3) Wafer Roughness

Lot No.	Spec	Sample Size	Test Step	Result	Data (nm)					Min.	Max.	Avg.	Std.	Cpk.
DOE Lot	Polishing Skip ≤ 200nm	5Points / Each Wafer	1차_1st wf	Pass/Fail	76.47	87.37	84.49	85.24	76.47	76.47	87.37	83.03	4.56	8.54
			1차_2nd wf	Pass/Fail	72.51	74.72	76.54	77.96	70.37	72.51	77.72	74.42	4.01	20.51
			2차_1st wf	Pass/Fail	73.80	82.70	77.34	71.60	66.52	66.52	82.70	74.39	2.04	8.68
			2차_2nd wf	Pass/Fail	75.06	83.20	74.09	72.53	83.75	72.53	83.75	80.44	4.29	3.29
			3차_1st wf	Pass/Fail	74.70	74.70	74.76	76.57	73.01	73.01	76.57	79.18	2.45	10.05
			3차_2nd wf	Pass/Fail	72.70	70.29	74.42	74.42	81.38	70.29	81.38	76.31	1.71	17.62

SATURN Project _ DOE Report

1. Back Grind Process Condition & Result

4) Wafer Thickness

Lot No	Spec	Sample Size	Test Step	Result	Data														Min.	Max.	Avg.	Std.	Cpk.
DOE Lot	250±10um (Target : 250um)	13Points/ Each Wafer	1차_1st wf	Pass	249.1	252.2	249.5	251.6	248.7	250.2	248.6	249.3	252.8	250	250.5	248.6	252.9	248.6	252.9	250.36	1.52	2.11	
			1차_2nd wf	Pass	249.6	250.4	251.1	249.8	251.3	251.8	251.6	249.2	250	249.5	249.1	251.1	249.9	248.3	251.4	250.58	0.9	2.81	
			2차_1st wf	Pass	249.4	249.4	252.1	251.6	250.1	250.7	250.1	249.3	249.1	250.8	249.1	249.2	248.6	242.5	250.9	250.38	1.12	3.23	
			2차_2nd wf	Pass	252	250.1	250.8	250.3	251.1	249.4	250	249.6	252.2	250.5	250.3	249.7	249.4	249.3	252	249.87	0.49	6.66	
			3차_1st wf	Pass	251.3	249.2	250.7	249.8	250.5	250.7	250.4	250.4	250.1	251.5	251.8	248	251.4	248	252.2	250.62	0.76	4.13	
			3차_2nd wf	Pass	249.7	250	251.1	251.9	253	251.5	251.3	250.9	250.3	249.7	251.5	253	250	249.7	253	251.1	1.1	1.7	

5) Wafer TTV (Total Thickness Variation) - Wafer 1매의 두께 오차(평탄도)

Lot No	Spec	Test Step	Inspection Results	
DOE Lot	12inch : [Max-Min] ≤ 10um	1차_1st wf	Pass	4.3 um
		1차_2nd wf	Pass	3.1 um
		2차_1st wf	Pass	8.4 um
		2차_2nd wf	Pass	2.7 um
		3차_1st wf	Pass	4.2 um
		3차_2nd wf	Pass	3.3 um

SATURN Project _ DOE Report

2. Saw Process Condition & Result

1) Process Condition

Item	Spec	Applied Condition	Remark
Cut Method	Wafer 특성에 따름	Step Cut	
Blade Model	Wafer 특성에 따름	Z1 : SD3500-N1-50-CE Z2 : SD3500-N1-50-FD	
Cut Speed (Step Cut)	Max 80mm/sec	60 mm/sec	규격 만족
Spindle RPM	30000~55000 RPM	Z1 : 50,000 / Z2 : 40,000 RPM	규격 만족
Blade Height (Z1 / Z2)	Wafer + Mount Tape 특성에 따름	Z1 : 0.240 mm / Z2 : 0.080 mm	

2) Visual Inspection Result (9Page 참조 / Visual 결과는 특이사항 없는 것으로 함)

Item	Lot No	Sample Size	Inspection Results	
Visual Inspection	DOE Lot	2 wafer / each evaluation	Pass / Fail	2 wafer / each evaluation

3) Kerf Width

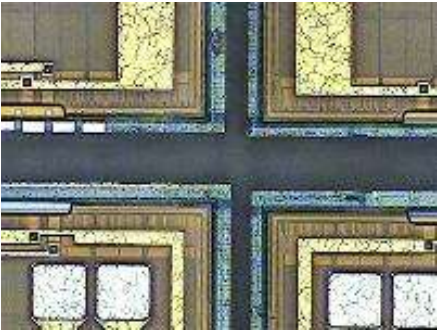
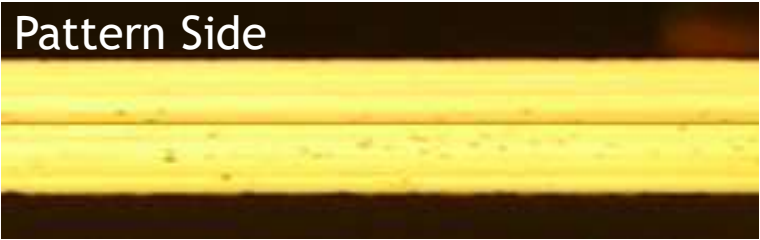
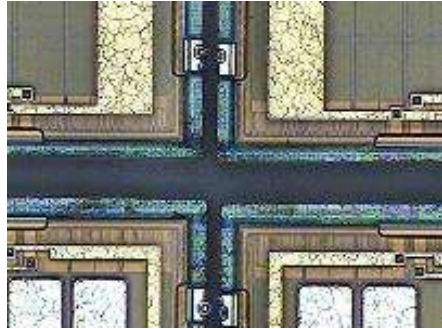
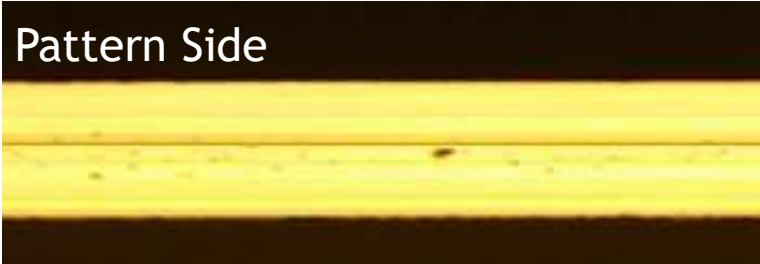
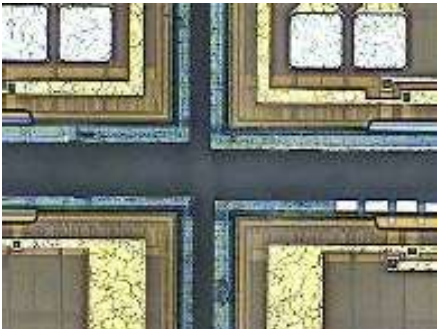
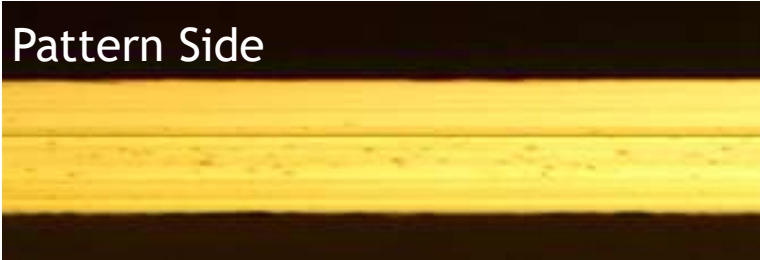
Lot No	Spec	Sample Size	Test Step	Result	Data										Min.	Max.	Avg.	Std.	Cpk.
DOE Lot	Max 65um	10Points/ Each Wafer	1차_1st wf	Pass/Fail	64.9	63.3	62.3	61.8	63.5	60.5	59.8	60.5	64.3	62.5	59.8	64.9	62.34	1.71	0.52
			1차_2nd wf	Pass/Fail	60.2	60.9	60.4	60.1	60.9	63.3	59.4	59.3	64.3	61.6	59.3	64.3	61.04	1.61	0.81
			2차_1st wf	Pass/Fail	59.8	59.2	62.7	62.4	59.4	60.4	63.9	63.8	62.3	59.5	59.2	63.8	61.34	1.87	0.65
			2차_2nd wf	Pass/Fail	61.2	60.5	60.8	59.5	61.5	60.1	59.5	62.3	62.8	63.6	59.5	63.6	61.18	1.38	0.92
			3차_1st wf	Pass/Fail	63	60.6	59.8	62.3	62.5	60	59.9	62.5	61.1	61	59.8	63	61.27	1.21	1.02
			3차_2nd wf	Pass/Fail	60.3	64.2	63.4	61.1	60.9	64.6	61.3	61.7	61.8	59	59	64.6	61.83	1.76	0.6

SATURN Project _ DOE Report

3. Saw Process Result

5) Image

- There is no top or back side chipping, lateral crack. Sawing condition is all ok.

Sample	Result	Top Image	Side Image
1 st	Pass		
2 nd	Pass		
3 rd	Pass		

※ 주어진 상황 중 오류 발견하기

- 앞에서 작성하신 DOE Report 를 자세히 훑어보고 멘토가 제공하여 드린 Data 중 Engineer Evaluation 결과로 적합하지 않은 부분을 찾고 그 이유를 기술해 주세요.

첫번째 발견된 문제점 : Back Grind 두께 데이터 중 2차_1st Wafer의 2번 측정 포인트 값이 222um로 기재

부적합 이유

1. SATURN 웨이퍼 두께 Spec은 $250 \pm 10 \text{ um}$ (240 ~ 260 um)이므로, 222 um는 규격 하한을 크게 벗어난 심각한 Spec-out(불량)임.
2. 그러나 동일 열의 최소값(Min)에는 242.5 um로 잘못 기재되어 있고 판정도 'Pass' 처리되어 있음.
3. 이는 장비 헌팅 에러 또는 단순 오기입(242.x um 오타)으로 보이며, 이상치를 확인/재측정하지 않고 그대로 통계 처리하여 'Pass' 낸 것은 엔지니어링 데이터로 부적합함.

두 번째 발견된 문제점 : Saw 공정 Kerf Width 결과가 1~3차 모두 'Pass'로 판정되어 있으나, 산출된 공정 능력지수(Cpk)는 0.52 ~ 1.02에 불과함.

부적합 이유

1. 반도체 양산 승인(Mass Qual) 기준 최소 요구 Cpk는 통상 1.33 이상 (우수 공정은 1.67 이상)임.
2. 실측치 자체는 상한 규격(Max 65 um) 미만이나, 평균값(61~62 um)이 상한 규격에 너무 근접하여 공정 마진이 매우 부족함.
3. 따라서 Cpk 기준(1.33)을 미달함에도 단순 실측 수치가 규격 내에 있다는 이유로 'Pass' 판정하여 양산으로 넘기는 것은 엔지니어 평가 결과로 적합하지 않음.